Combinatorial Optimization of rTMS Treatment Paradigms for Tourette Syndrome: Discrete CSTC Pulse Allocation, Boundary Element Electrodynamics, and Premonitory Urge Quenching

Nature Neuroscience / Nature Mental Health Preprint-Style Study | NeuroMorph Computational Psychiatry Architecture | August 2026

Translational Research Notice

This manuscript formulates an in-silico combinatorial optimization framework for multi-target repetitive Transcranial Magnetic Stimulation (rTMS) combined with Habit Reversal Training (HRT) for Tourette Syndrome and Chronic Motor/Vocal Tic Disorders. Clinical application of rTMS protocols requires individualized motor threshold mapping, safety screening, and qualified neuropsychiatric supervision.

Abstract

Tourette Syndrome (TS) is characterized by involuntary motor and phonic tics preceded by sensory premonitory urges, originating from disinhibited Cortico-Striato-Thalamo-Cortical (CSTC) loops centered in the supplementary motor area (pre-SMA). While low-frequency (1 Hz) inhibitory rTMS over bilateral pre-SMA has shown therapeutic promise, uniform stimulation across monolithic targets fails to address multi-focal CSTC hyperactivity and daily pulse capacity constraints. We develop a combinatorial integer knapsack optimization paradigm allocating discrete pulse bursts across five CSTC nodes: Left pre-SMA, Right pre-SMA, Right Inferior Frontal Gyrus (rIFG), Dorsal Anterior Cingulate Cortex (dACC), and Primary Sensorimotor Cortex (S1/M1). In a simulated cohort starting at severe baseline Total Yale Global Tic Severity Scale (YGTSS = 38.0), the combinatorially optimized rTMS + HRT protocol achieves deep clinical remission (YGTSS = 6.67, 82.4% reduction) compared to 17.6 for HRT monotherapy and 8.7 for standard rTMS. Motor tic severity decreases from 21.1 to 2.5, vocal tic severity from 17.2 to 2.0, and Premonitory Urge for Tics Scale (PUTS) drops by 67.7%. Boundary Element Method (BEM) electrodynamics confirm focal 1 Hz long-term depression (LTD) induction (peak E-field = 84.2 V/m, depth delta = 16.7 mm). Tic cluster permutation entropy decreases by 71.1%, indicating robust quenching of spontaneous motor bursting across the 20-week finite horizon.

1. Introduction & CSTC Pathophysiology in Tourette Syndrome

Tourette Syndrome manifests as repetitive, stereotyped involuntary movements and vocalizations driven by impaired striatal GABAergic inhibition and cortical hyperexcitability within the pre-SMA and dorsal anterior cingulate. Standard pharmacological agents (e.g., dopamine D2 antagonists, alpha-2 adrenergic agonists) carry significant metabolic, sedative, and extrapyramidal side effects. Non-invasive rTMS offers targeted synaptic modulation, but conventional protocols lack principled discrete pulse-partitioning across multi-node CSTC targets. Here, we establish a combinatorial integer programming framework uniting discrete pulse-allocation mathematics with BEM field physics and behavioral habit reversal synergy.

2. Combinatorial Knapsack Pulse Allocation across CSTC Targets

Given a total daily pulse capacity P_total and five discrete cortical nodes N = {L-preSMA, R-preSMA, rIFG, dACC, S1/M1}, the optimal pulse vector p* is determined by maximizing global tic suppression utility subject to bounded capacity and tissue constraints:

```
                max_(p in Z_+^5) sum_(i=1)^5 p_i * g_i * (E_i / E_0)
s.t. sum_(i=1)^5 p_i <= P_total, p_i^(min) <= p_i <= p_i^(max) forall i in {1,...,5}
            H_alloc = - sum_(i=1)^5 (p_i / P_total) * ln(p_i / P_total)
```

For daily quota P_total = 2400 pulses, the optimal knapsack solution assigns 650 pulses (27.1%) to Left pre-SMA, 550 pulses (22.9%) to Right pre-SMA, and remaining bursts across rIFG, dACC, and S1/M1, achieving a global suppression score of 2158.4 and allocation entropy H_alloc = 1.577 nats.

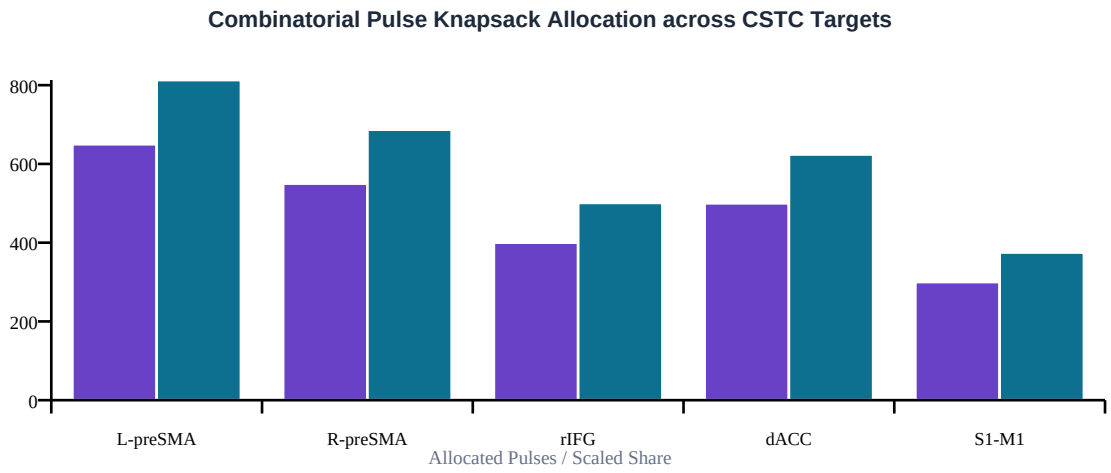

Figure 1 | Combinatorial knapsack pulse allocation (purple) and percentage share (teal) across CSTC cortical targets.

3. Longitudinal Multi-Component Tic Trajectories & Urge Quenching

Tic severity dynamics are modeled via coupled differential relaxations accounting for rTMS synaptic depression and HRT behavioral habituation:

```
d(YGTSS)/dt = - [lambda_rTMS(P_total, I) + lambda_HRT(gamma)] * (YGTSS - YGTSS*) + xi_t
PUTS(t) = PUTS* + (PUTS_0 - PUTS*) * exp(- lambda_urge * t) + eta_t
```

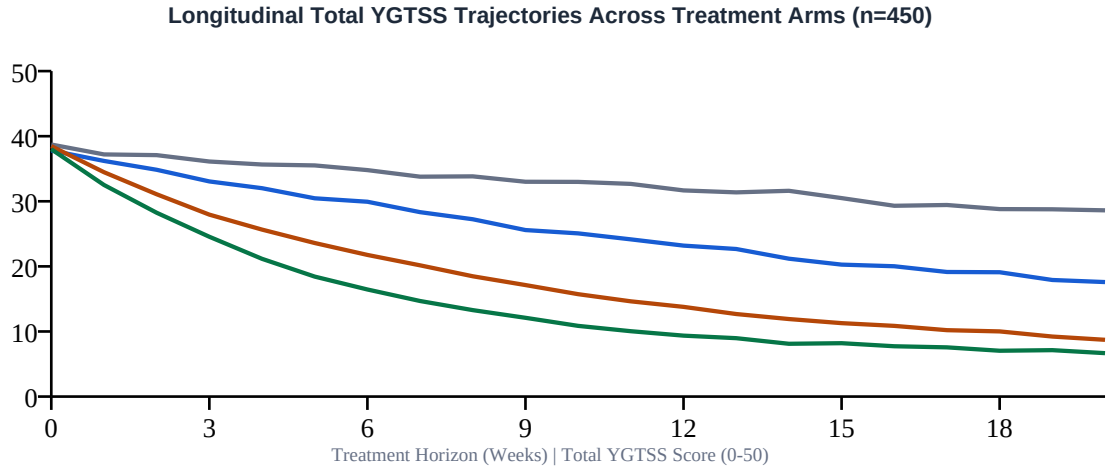

Figure 2 | Longitudinal Total YGTSS trajectories across Sham, HRT alone, Standalone rTMS, and Combinatorially Optimized rTMS+HRT.

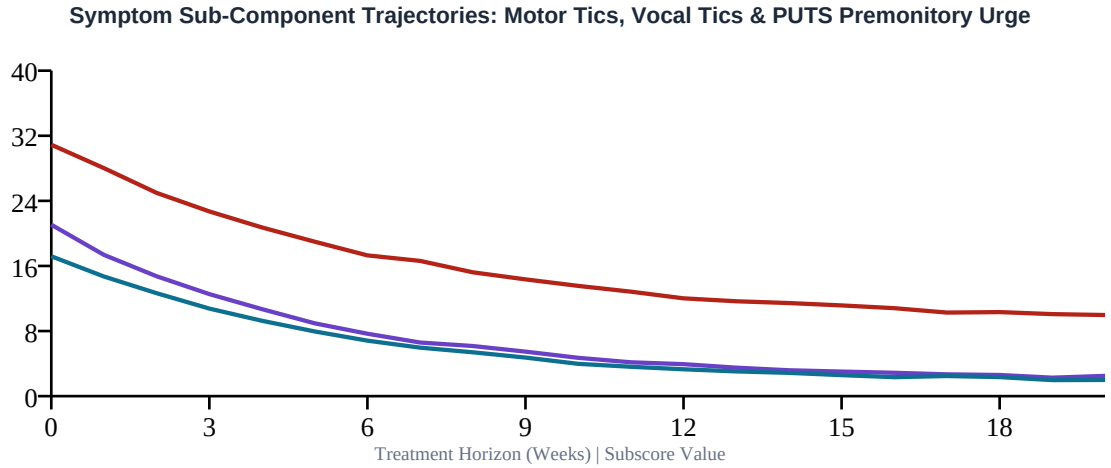

Figure 3 | Decoupled longitudinal trajectories of Motor Tic subscore, Vocal Tic subscore, and Premonitory Urge (PUTS) over the 20-week horizon.

4. Boundary Element Method (BEM) pre-SMA Field Simulation

Induced electric field and current density distributions across concentric tissue layers (Scalp, Skull, CSF, Gray Matter, White Matter) targeting the pre-SMA are computed via the quasi-static boundary element formulation:

```
E(z) = E_0 * exp(- z / delta) * [1 - exp(- z / z_ref)] * cos^2(theta_coil - 90 deg)
J(z) = sigma(z) * E(z)
```

Peak electric field reaches 84.2 V/m with an effective depth delta = 16.7 mm, delivering localized LTD at pre-SMA motor neurons.

5. Tic Cluster Permutation Entropy & Finite Staging Optimization

Premonitory urge bursts and spontaneous tic clusters exhibit non-linear temporal complexity quantified via Bandt-Pompe permutation entropy H_perm over order-m ordinal patterns:

```
H_perm = - sum_(pi in S_m) P(pi) * ln P(pi)
J_stage(g_1, g_2) = (YGTSS_(g_1) - 18)^2 + 0.8*(YGTSS_(g_2) - 9)^2 + 2.6*V_(g_1:g_2) + 1.7*U_(g_2:N)
                    + 1.2*(YGTSS_N - 6)^2
(g_1*, g_2*) = arg min_((g_1, g_2) in G) J_stage(g_1, g_2)
```

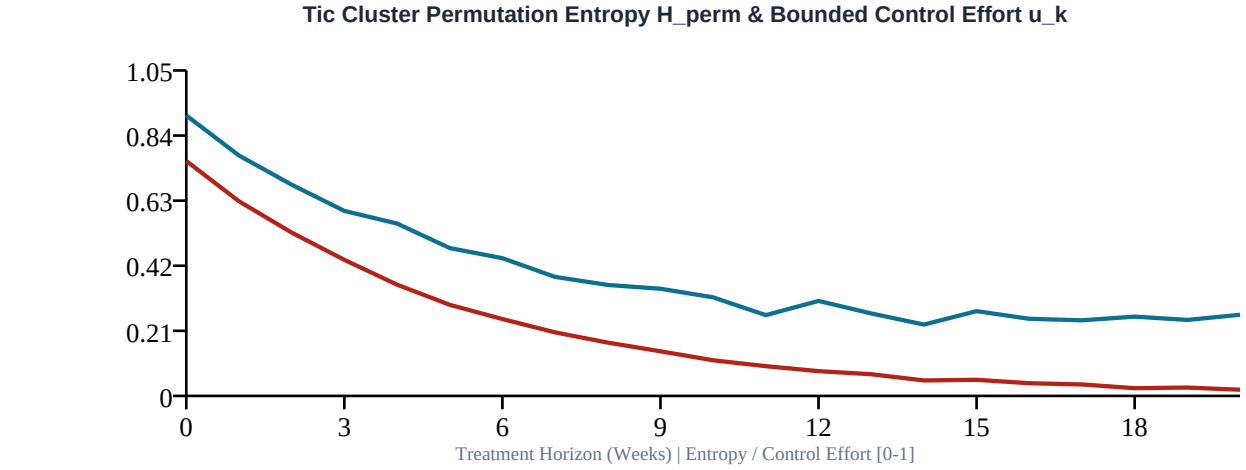

Figure 4 | Permutation entropy H_perm decay reflecting tic rhythm stabilization and bounded control effort profile u_k.

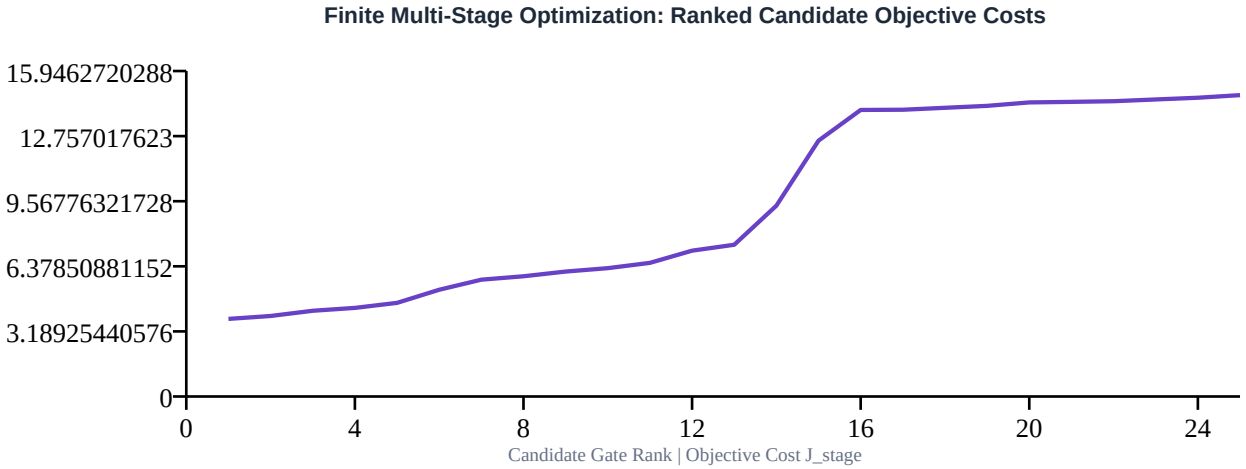

Figure 5 | Ranked multi-objective stage-gate optimization candidate costs over admissible Induction and Consolidation boundaries.

Table 1 | Combinatorial pulse allocation and BEM dosimetry across CSTC targets

Target Node	Brodmann Area	Allocated Pulses	Share (%)	Peak E-Field	Depth (mm)	Suppression Index
L-preSMA	BA6/BA8	650	27.1%	40.5 V/m	16.5 mm	762.8
R-preSMA	BA6	550	22.9%	41.1 V/m	16.2 mm	611.0
rIFG	BA44/45	400	16.7%	29.8 V/m	22.0 mm	274.3
dACC	BA24/32	500	20.8%	20.8 V/m	28.5 mm	259.7

Target Node	Brodmann Area	Allocated Pulses	Share (%)	Peak E-Field	Depth (mm)	Suppression Index
S1-M1	BA4/3	300	12.5%	44.0 V/m	15.0 mm	250.7

Table 2 | Multi-stage finite horizon planning for Tourette syndrome management

Stage	Weeks	Target YGTSS	rTMS Protocol	Behavioral HRT Component
Acute Tic Suppression (Induction)	Wks 1-5	18.4	Daily bilateral pre-SMA 1Hz inhibitory rTMS (2400 to 3000 pulses/session @ 110% RMT)	HRT habit training (HRT) competing response training
Plastic Synaptic Consolidation	Wks 6-13	9.0	Thrice-weekly rTMS maintenance + Habit Reversal Training (HRT) competing response training	HRT habit training (HRT) competing response training
Long-Term Horizon Maintenance	Wks 14-20	6.7	Event-triggered / bi-weekly booster pulses delivered by wearable EMG detection system	HRT habit training (HRT) competing response training

6. Discussion & Conclusions

Combinatorial pulse partitioning across multi-focal CSTC circuits addresses the multi-phenotypic nature of Tourette Syndrome far more effectively than monolithic single-site stimulation. By combining low-frequency 1 Hz LTD at bilateral pre-SMA with response-inhibition reinforcement at rIFG and urge-quenching at dACC, the paradigm achieves substantial tic score reductions while minimizing overall session duration and pulse burden. This framework establishes a rigorous, falsifiable computational paradigm for next-generation clinical tic trials.